Baseline Notebook

Setup Environment

```
# DO NOT MODIFY THE CODE IN THIS CELL
!pip install -q utstd

from utstd.folders import *
from utstd.ipyrenders import *

at = AtFolder(
    course_code=36106,
    assignment="AT3",
)
at.run()

import warnings
warnings.simplefilter(action='ignore')
```

```
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0:00:00
━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.9/4.9 MB 48.2 MB/s eta
0:00:00
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
umap-learn 0.5.12 requires scikit-learn>=1.6, but you have scikit-
learn 1.5.2 which is incompatible.
hdbscan 0.8.42 requires scikit-learn>=1.6, but you have scikit-learn
1.5.2 which is incompatible.
Mounted at /content/gdrive

You can now save your data files in:
/content/gdrive/MyDrive/36106/assignment/AT3/data
```

Student Information

```
group_name = "Group 20"
student_name = "Ratnadeep Patra"
student_id = "26294153"

# Do not modify this code
print_tile(size="h1", key='group_name', value=group_name)
```

```
<IPython.core.display.HTML object>
```

```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h1", key='student_name', value=student_name)
```

```
<IPython.core.display.HTML object>
```

```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h1", key='student_id', value=student_id)
```

```
<IPython.core.display.HTML object>
```

---

0. Python Packages

0.a Install Additional Packages

> If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

```
# <Student to fill this section and then remove this comment>
```

0.b Import Packages

```
# <Student to fill this section and then remove this comment>
import pandas as pd
import altair as alt
```

A. Assess Baseline Model

```
# DO NOT MODIFY THE CODE IN THIS CELL
# Load data
try:
  X_train = pd.read_csv(at.folder_path / 'X_train.csv')
  y_train = pd.read_csv(at.folder_path / 'y_train.csv')

  X_val = pd.read_csv(at.folder_path / 'X_val.csv')
  y_val = pd.read_csv(at.folder_path / 'y_val.csv')

  X_test = pd.read_csv(at.folder_path / 'X_test.csv')
  y_test = pd.read_csv(at.folder_path / 'y_test.csv')
except Exception as e:
  print(e)
```

```
[Errno 2] No such file or directory:
'/content/gdrive/MyDrive/36106/assignment/AT3/data/y_train.csv'
```

A.1 Generate Predictions with Baseline Model

```
# In anomaly detection, a true baseline such as a dummy classifier or
dummy regressor generally does not exist because there are no reliable
ground-truth
# labels defining what is correct or incorrect. In supervised learning
tasks like classification and regression, baselines work because
predictions can be
# directly compared against known target labels or values using simple
strategies such as predicting the majority class or mean value.
However, anomaly
# detection is unsupervised, where the model itself identifies unusual
patterns from the data distribution. Since anomalies are rare,
ambiguous, and
# context-dependent, a trivial predictor such as "everything is
normal" does not provide a meaningful comparison. Without labeled
anomalies, there is no
# objective way to evaluate whether a simplistic detector is
performing poorly or whether the detected anomalies are valid.
Therefore, traditional baseline
# analysis is skipped for anomaly detection.
```

A.2 Selection of Performance Metrics

Provide some explanations on why you believe the performance metrics you chose is appropriate

```
# No metric needs to be considered

performance_metrics_explanations = """
In unsupervised anomaly detection, there is no definitive performance
metric because no ground-truth target labels are available to verify
whether each
flagged record is truly anomalous or normal. Unlike supervised
learning tasks, the model predictions cannot be directly compared
against a known target
outcome, so standard metrics such as accuracy, precision, recall or
F1-score are not appropriate. Therefore, the baseline model cannot be
evaluated using
traditional predictive-performance measures.

However, this does not mean that the model cannot be assessed at all.
Instead, the evaluation focuses on unsupervised and business-oriented
indicators
such as anomaly count, anomaly rate, LOF score distribution and score
separation between normal and anomalous transactions. These indicators
do not prove
that every flagged record is a true anomaly, but they help judge
whether the model is producing a manageable, interpretable and
meaningful anomaly shortlist
```

```
for business investigation. For this reason, the final LOF model
(discussed in the anomaly notebook - 36106-26AU-AT3-26294153-SC-c-
anomaly.ipynb) is
evaluated using LOF score behaviour, anomaly count and separation
between normal and anomalous groups, rather than supervised
performance metrics.
"""

# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='performance_metrics_explanations',
value=performance_metrics_explanations)
```

```
<IPython.core.display.HTML object>
```

A.3 Baseline Model Performance

Provide some explanations on model performance

```
# No baseline analysis and performance metric to get baseline model
performance

baseline_performance_explanations = """
As noted in the baseline model prediction and performance metric
selection, a traditional baseline model and its performance metrics
are not applicable in
unsupervised anomaly detection. The absence of ground-truth labels
makes it impossible to objectively quantify the performance of a
'baseline' that merely
predicts normalcy or random anomalies. Therefore, a quantitative
baseline performance evaluation is not provided here.
"""

# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='baseline_performance_explanations',
value=baseline_performance_explanations)
```

```
<IPython.core.display.HTML object>
```